UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	INTRODUCTION TO INTERACTIVE MULTIMEDIA
COURSE CODE	CSC413
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

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PART A (15 MARKS)

State TRUE (T) or FALSE (F) for the following statements.

1. Card-based authoring tools are best for messages with a clear beginning and end.
2. A prototype is an early model of a product.
3. CMYK is used for printed output.
4. A hyperlink can link only to another web page, not to a file.
5. Interactivity gives users control over content and flow.
6. A linear presentation allows users to choose their own path.
7. Higher resolution always means a smaller file.
8. A scanner is an input device.
9. A storyboard is a visual plan of a multimedia sequence.
10. Sampling converts a continuous sound wave into digital values.
11. A bitmap image is made up of pixels.
12. Chroma key compositing replaces a colour background with another image.
13. A low-fidelity prototype is used to test the basic layout and flow.
14. Hypermedia combines multimedia with hyperlinks.
15. Tweening creates intermediate frames between keyframes.

PART B (15 MARKS)

Select the most appropriate answer from Table A for each of the following questions.

Table A

A. Hypermedia	B. Bitmap	C. Plotter	D. Sampling	E. Vector
F. MIDI	G. Keyframe	H. Frame rate	I. Multimedia	J. Virtual reality
K. Browser	L. Scanner	M. Interactivity	N. Linear multimedia	O. Tweening

1. integration of text, graphics, audio, video and animation
2. converting sound into digital values
3. input device that digitises documents
4. frames displayed per second
5. simulated environment
6. presentation without user control of sequence
7. image made of pixels
8. linked multimedia elements
9. image made of mathematical objects
10. software that displays web pages
11. user control over content and flow
12. standard for musical notes and instructions
13. important frame in an animation
14. generating in-between frames
15. output device for large drawings

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PART C (30 MARKS)

Choose the correct answer for each question.

1. Hertz measures:
A. Frequency
B. Loudness
C. File size
D. Resolution

2. A storyboard is:
A. A visual sequence plan
B. A video codec
C. A sound file
D. A colour model

3. A higher sampling rate generally:
A. Changes colour
B. Removes audio
C. Reduces accuracy
D. Improves accuracy

4. MPEG is:
A. A video compression family
B. A storage device
C. A font
D. An image editor

5. Sampling converts:
A. Digital video to text
B. Text to speech
C. Analog sound to digital values
D. Images to vectors

6. Interlacing:
A. Adds sound
B. Draws every line at once
C. Draws alternate lines of a frame
D. Removes colour

7. HTML is used to:
A. Structure web content
B. Model 3D objects
C. Compress video
D. Store audio samples

8. Which is a benefit of multimedia in education?
A. It removes all interaction
B. It replaces all teachers
C. It combines visuals and sound
D. It uses only text

9. Streaming allows media to:
A. Require full download first
B. Play while downloading
C. Only play offline
D. Remove compression

10. Plug and play means:
A. No power needed
B. Manual driver installation only
C. Automatic device recognition
D. Wireless only

11. Which is an example of linear multimedia?
A. An interactive kiosk
B. A game with choices
C. A website with menus
D. A cinema presentation

12. Path animation moves an object:
A. Randomly
B. Only vertically
C. Without motion
D. Along a predefined route

13. Morphing changes:
A. Audio pitch
B. Font size
C. One shape into another
D. File format

14. Hypertext contains:
A. Links to other content
B. Only images
C. Only audio
D. No text

15. A hyperlink links to:
A. Only images
B. Only text
C. Only audio
D. Another resource

16. A serif font has:
A. No strokes
B. No spacing
C. Only uppercase letters
D. Small strokes at character ends

17. Which is an output device?

A. Keyboard
B. Mouse
C. Scanner
D. Plotter

18. Which image type scales without losing quality?

A. Photograph
B. Scan
C. Bitmap
D. Vector

19. Anti-aliasing is used to:

A. Increase file size only
B. Compress video
C. Smooth jagged edges
D. Add sound

20. Frame rate is:

A. Bits per second
B. Frames per second
C. Samples per second
D. Pixels per inch

21. Which is an input device?

A. Printer
B. Plotter
C. Monitor
D. Scanner

22. Kerning adjusts:

A. Font colour
B. Space between individual character pairs
C. Line length
D. Page size

23. Which is NOT a multimedia element?

A. Compiler
B. Text
C. Audio
D. Animation

24. Which is an authoring tool category?

A. RGB
B. JPEG
C. Card-based
D. CMYK

25. A bitmap image is made of:

A. Pixels
B. Text characters
C. Sound samples
D. Mathematical objects

26. A prototype is:

A. A final release
B. A font
C. An early model of a product
D. A codec

27. Cel animation is created by:

A. Recording sound
B. Drawing each frame by hand
C. Scanning text
D. Printing images

28. A licence gives:

A. Unlimited bandwidth
B. Free hardware
C. Ownership of all works
D. Permission to use a work

29. An authoring tool is used to:

A. Only browse the web
B. Only print text
C. Assemble multimedia elements
D. Only record audio

30. Keyframes are:

A. Colour profiles
B. Important frames in an animation
C. Audio samples
D. Web links

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PART D (40 MARKS)

QUESTION 1

A museum wants a website that presents its collection and lets visitors explore related exhibits.

a) State TWO differences between linear and non-linear navigation.

(2 marks)

b) Describe TWO roles of hyperlinks in the website.

(4 marks)

c) Explain how streaming and compression affect multimedia delivery on the web.

(4 marks)

QUESTION 2

A shopping complex requires a kiosk that gives customers location and retailer information.

a) State TWO important project team roles for completing the kiosk.

(2 marks)

b) Design a low-fidelity prototype for a pre-school courseware home page. The page must show topic menus, the course title and the school logo.

(3 marks)

c) Discuss TWO factors that lead to the wide use of mobile devices in games and shopping applications.

(5 marks)

QUESTION 3

Text is an essential element in a multimedia project. Its shape, size, design, colour, arrangement and linking all affect the message.

a) State TWO reasons sans serif is mostly used for screen display and serif is used for printed material.

(3 marks)

b) Describe TWO roles of hypertext in multimedia.

(3 marks)

c) Explain how text can have a massive impact on an informative multimedia project in the link and HTML context.

(4 marks)

QUESTION 4

A company wants an interactive product demonstration that runs on a kiosk and on the web.

a) State TWO features to consider when choosing authoring software.

(3 marks)

b) Explain the interactivity level suitable for the demonstration.

(3 marks)

c) Explain TWO authoring tool categories appropriate for this project.

(4 marks)

END OF QUESTION PAPER